

A VISUAL FIELD GUIDE FOR BUSINESS ANALYSIS

The Measurement Atlas

Meaning. Method. Visual evidence.

01

Nominal

Identity

02

Ordinal

Order

03

Interval

Difference

04

Ratio

Relative size

16 measure subtypes | 16 directed combinations

20 chart families | 8 worked business examples

All 14 GC1015 sessions + an optional path beyond

Professor Ken Ezekwem

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Optional learning reference. Complete during or after the course. This guide does not add assessed requirements or change rules for exams and independent work.

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Read this as a field guide: define the question and grain, inspect the measure subtype, choose a method family, then check the design and communicate evidence. All numerical examples are synthetic and distinct from assessed responses.

[Open the interactive Measurement Atlas](#)

01 / FIELD GUIDE

Before choosing a method

A measurement level describes meaning. It does not, by itself, determine a test. The outcome, estimand, distribution, dependence and study design complete the decision.

Question

Describe, compare, associate, predict or explain?

Name the population and the quantity you want to estimate. A causal claim needs a defensible design, not a correlation alone.

Grain

What does one row represent?

One customer, order, visit, store-day or survey response? Aggregating orders to stores changes both the variable and the unit of analysis.

Meaning

What information do the values contain?

Record level, units, valid categories, allowable range, true zero, and whether values are discrete, bounded or circular.

Design

Which rows depend on each other?

Independent groups, paired units, repeated observations, clusters and time dependence lead to different standard errors and methods.

Evidence

What must be checked?

Audit missing values, duplicates, denominators, outliers and selection. Inspect distributions and model residuals before trusting inference.

Communication

What would change a decision?

Report effect size, interval, sample/exposure size, practical threshold, key limits and a chart answering the question.

Sources: [Velleman & Wilkinson \(1993\)](#); [UCLA OARC \(n.d.\)](#)

Use a stronger measurement representation when supported, but do not confuse detail with quality. A precisely measured biased sample can still answer the wrong question.

02 / FIELD GUIDE

The four measurement levels

The traditional scale framework is useful for thinking about meaningful comparisons and recodings. Treat it as a starting point, not a rigid menu of permissible tests.

Nominal / Identity

Are these the same kind?

Labels distinguish categories. No intrinsic ranking is required.

Example: Sales channel: web, store, partner.

Summarize: Equality and membership; frequencies, proportions and mode.

Visual: Bar or dot chart; contingency table for two categories.

Preserving the scale: Any one-to-one relabeling preserves the information.

Watch for: A mean of channel codes (1, 2, 3) has no stable meaning.

Sources: [NIST \(2016\)](#); [Stevens \(1946\)](#)

Ordinal / Order

Which comes before which?

Values have a meaningful order, but adjacent gaps need not be equal.

Example: Satisfaction: very dissatisfied to very satisfied.

Summarize: Order, median category, quantiles and cumulative percentages. Preserve category labels when reporting the median.

Visual: Ordered bars or a diverging stacked bar for survey responses.

Preserving the scale: Any strictly increasing recoding preserves the order.

Watch for: A rating of 4 is not demonstrably twice the satisfaction of 2.

Sources: [NIST \(2016\)](#); [Liddell & Kruschke \(2018\)](#)

Interval / Difference

How far apart are they?

Equal increments represent equal differences. Zero is a chosen reference.

Example: Temperature in °C; timestamps from a specified epoch.

Summarize: Differences, arithmetic means and standard deviations, when the numerical model is meaningful.

Visual: Histogram, ECDF, box plot or scatterplot; ordered time on a line.

Preserving the scale: Positive affine transformation: $\text{new} = a \times \text{old} + b$, with $a > 0$.

Watch for: 20 °C is not twice as hot as 10 °C. Ratios of differences can be meaningful.

Sources: [NIST \(2016\)](#)

Ratio / Relative size

How many times as much?

Equal increments and a meaningful zero permit comparisons by ratio.

Example: Delivery duration in minutes; units sold; mass in kilograms.

Summarize: Differences and ratios. Mean, median and spread still depend on the question and distribution.

Visual: Histogram, ECDF, box plot or scatterplot; counts may use discrete bars.

Preserving the scale: Positive change of units: $\text{new} = a \times \text{old}$, with $a > 0$.

Watch for: A ratio scale does not imply normal data or justify every statistical test.

Sources: [NIST \(2016\)](#); [Velleman & Wilkinson \(1993\)](#)

Nominal and ordinal describe categorical information. Interval and ratio describe numerical comparisons. Discrete/continuous is a separate dimension: a count can be discrete and ratio; a recorded continuous amount can be rounded. Software storage types do not determine statistical meaning.

Sources: [NIST \(2016\)](#); [Velleman & Wilkinson \(1993\)](#)

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Decompose the measure

“Decompose” can mean unpacking a construct, a KPI formula, subgroup structure, a time series, or a multivariate signal. Keep those operations distinct.

From construct to recorded value

Service quality → satisfaction item → ordered response → stored code.

A concept is not its column. Define the instrument, wording, scoring and missing-data rules. Recode labels without claiming new measurement precision.

From KPI to numerator and denominator

Conversion = purchases / eligible visits; total rate = Σ purchases / Σ eligible visits.

Record eligibility, period, grain, exclusions and repeat visitors. The overall rate is a denominator-weighted average of subgroup rates, not usually their simple average.

From KPI to a product of drivers

Revenue = customers × orders per customer × revenue per order.

This is an accounting identity when all components use compatible populations and periods. It is not proof that any driver caused revenue. Zero denominators and changing definitions can break the calculation.

From total to additive parts

Revenue = web revenue + store revenue + partner revenue.

Parts must be mutually exclusive and exhaustive. A waterfall can show signed contributions that reconcile to a total; a treemap shows a positive hierarchy.

From aggregate to subgroup evidence

Overall purchase rate → rate within each device or customer segment.

Compare denominators and segment mix. Aggregated trends may reverse within subgroups (Simpson's paradox). Keep the original grain and avoid an ecological claim about individuals.

From time series to components

Observed value = trend-cycle + seasonal component + remainder.

An additive decomposition suits roughly constant seasonal magnitude; a multiplicative form suits scale-dependent seasonal magnitude for positive data. Decomposition is descriptive, not causal. Forecasts require separate validation.

From many variables to latent structure

Several standardized numerical measures → principal components.

PCA finds variance directions, not automatically a meaningful business construct. Scaling matters; variance explained is not predictive accuracy. Factor analysis has a different latent-variable model.

Sources: Hyndman & Athanasopoulos (2021); scikit-learn developers (n.d.); Velleman & Wilkinson (1993)

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Useful subtypes and special cases

Annotate the scale with support (possible values), geometry, construction and role. These distinctions often matter more for model choice than the interval-versus-ratio distinction.

Binary indicator

Nominal; two states

Example: 0/1 purchase flag

The coding mean equals the fraction of 1s. Model an individual event with Bernoulli/logistic methods; report which state is 1.

Method / visual: Proportion with interval; binary model

Sources: [statsmodels developers \(n.d.\)](#); [statsmodels developers \(n.d.\)](#)

Unordered multicategory

Nominal; three or more states

Example: Customer acquisition channel

Use indicator variables with a stated reference category. A numeric code is a label, not a distance.

Method / visual: Counts, cross-tabs, multinomial model

Sources: [UCLA OARC \(n.d.\)](#)

Ordered categories

Ordinal; coarse grades

Example: Low, medium, high risk

Preserve order. Report the complete distribution, not only a middle category. An ordinal predictor can use indicators unless a trend score is justified.

Method / visual: Ordered bars, cumulative logit

Sources: [statsmodels developers \(n.d.\)](#)

Ranks

Ordinal; relative position

Example: Supplier ranked 1st to 10th

Rank 1 may mean best or smallest; document direction and ties. Rankings do not measure the gap in performance.

Method / visual: Spearman/Kendall association

Sources: [SciPy developers \(n.d.\)](#)

Single Likert item

Ordinal; response scale

Example: One 1-to-5 satisfaction question

Numeric labels do not establish equal psychological gaps. Use category frequencies and ordinal/rank methods as a clear starting point.

Method / visual: Diverging bars, ordinal model

Sources: [Liddell & Kruschke \(2018\)](#)

Multi-item score

Construct requiring validation

Example: Mean of several service questions

A composite is not automatically interval. Check scoring, reverse-coded items, missing-item rules, dimensionality and reliability. If treating it as approximately continuous, explain and compare an ordinal or robust analysis.

Method / visual: Distribution plus sensitivity analysis

Sources: [Liddell & Kruschke \(2018\)](#); [Velleman & Wilkinson \(1993\)](#)

Count

Discrete, nonnegative; often ratio

Example: Tickets arriving per hour

Whole events, with a defined observation window. Poisson is a candidate, not guaranteed; inspect overdispersion and excess zeros.

Method / visual: Count bars; Poisson/negative binomial

Sources: [statsmodels developers \(n.d.\)](#)

Positive amount

Continuous ratio in a fixed unit

Example: Transaction amount or waiting duration

Often skewed. Gamma or lognormal models may suit strictly positive data. Zeros require a deliberate alternative; do not add a constant merely to take logs.

Method / visual: Histogram/ECDF; log scale where useful

Sources: [statsmodels developers \(n.d.\)](#)

Signed balance or change

Numerical difference; zero denotes balance

Example: Net profit or change in minutes

Negative values matter. Do not assume “twice as much” is a useful business interpretation across zero. Ordinary logs cannot represent negatives.

Method / visual: Diverging bars, waterfall, linear model

Sources: [Velleman & Wilkinson \(1993\)](#)

Counted fraction

Successes / known trials

Example: 24 purchases / 200 eligible visits

Retain numerator and denominator. 12% from 20 visits and 12% from 2,000 visits have different precision. Account for repeated visitors or clustering.

Method / visual: Binomial interval/model

Sources: [statsmodels developers \(n.d.\)](#); [statsmodels developers \(n.d.\)](#)

Continuous proportion

Bounded numerical value

Example: Fraction of each workday spent on analysis

Different from successes out of known trials. Beta regression is a candidate for values strictly between 0 and 1; exact endpoints need an appropriate extension or fractional-response model.

Method / visual: Bounded distribution, beta/fractional model

Sources: [Cribari-Neto & Zeileis \(2010\)](#)

Rate or derived ratio

Numerical value with an exposure unit

Example: Tickets per 100 operating hours

Preserve the exposure and compare like denominators. A count model can use $\log(\text{exposure})$ as an offset; a rate is not necessarily bounded by 1.

Method / visual: Rate dot plot with intervals

Sources: [statsmodels developers \(n.d.\)](#)

Calendar time / duration

Timestamp: interval; elapsed time: ratio

Example: Order date versus fulfillment hours

Date differences yield durations after timezone and calendar conventions are defined. Time series may be dependent; a timestamp is not an ID.

Method / visual: Line, seasonal panels, duration ECDF

Sources: [Hyndman & Athanasopoulos \(2021\)](#)

Cyclic direction / clock time

Circular geometry beyond a linear scale

Example: Arrival hour; wind direction

23:00 and 01:00 are close across midnight. The arithmetic mean is noon, while their circular midpoint is midnight. A circular mean is undefined or unstable when directions cancel.

Method / visual: Circular summaries, polar histogram

Sources: [SciPy developers \(n.d.\)](#); [Chrisman \(1998\)](#)

Composition

Vector of constrained shares

Example: 40% labor, 35% materials, 25% logistics

Shares sum to a fixed total, so components cannot move independently. Use log-ratio methods for inference when appropriate; zeros need explicit treatment.

Method / visual: 100% stacked bars, ternary diagram

Sources: [van den Boogaart & Tolosana-Delgado \(n.d.\)](#)

Identifier / text / location

Role-specific representations

Example: Order ID, feedback text, latitude

IDs support joins and de-duplication, not average-ID analysis. Text needs documented coding or features. Maps require a spatial question; spatial dependence and projection matter.

Method / visual: Quality checks, coded frequencies, map

Sources: [Velleman & Wilkinson \(1993\)](#); [Holtz & Healy \(n.d.\)](#)

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Combinations of measurement levels

Direction matters: predicting a category from revenue needs a categorical outcome model; predicting revenue across categories needs a numerical outcome model. Association alone may be described symmetrically.

X below is the predictor/grouping variable. Y is the outcome. All 16 directed pairs are included. These are candidate families; the subtype and design override a mechanical scale-only choice.

X predictor	Y outcome	Candidate / visual
Nominal	Nominal	Cross-tab / categorical model Grouped proportions or predicted-category probabilities <i>Channel to purchase status</i>
Nominal	Ordinal	Ordered distribution / ordinal model Ordered category proportions by group <i>Channel to satisfaction</i>
Nominal	Interval	Group comparison / linear model Group dots/box plots and mean intervals <i>Facility to temperature (°C)</i>
Nominal	Ratio	Group comparison / linear model Group dots/box plots and mean intervals <i>Channel to transaction amount</i>
Ordinal	Nominal	Cross-tab / categorical model Grouped proportions or predicted-category probabilities <i>Service tier to channel choice</i>
Ordinal	Ordinal	Ordered distribution / ordinal model Rank association or predicted category probabilities across X <i>Service tier to satisfaction</i>
Ordinal	Interval	Rank trend / flexible numerical model Jittered ordered X with outcome distributions <i>Operating mode (low/medium/high) to temperature</i>
Ordinal	Ratio	Rank trend / flexible numerical model Jittered ordered X with outcome distributions <i>Service tier to delivery minutes</i>
Interval	Nominal	Logistic / multinomial model Outcome probabilities across X; numerical distribution by category <i>Temperature to failure status</i>
Interval	Ordinal	Ordered distribution / ordinal model Rank association or predicted category probabilities across X <i>Temperature to comfort rating</i>

X predictor	Y outcome	Candidate / visual
Interval	Interval	Scatter / correlation / regression Scatterplot, fitted relationship and residual checks <i>Ambient temperature to product temperature</i>
Interval	Ratio	Scatter / correlation / regression Scatterplot, fitted relationship and residual checks <i>Temperature to energy consumption</i>
Ratio	Nominal	Logistic / multinomial model Outcome probabilities across X; numerical distribution by category <i>Tenure in months to cancellation</i>
Ratio	Ordinal	Ordered distribution / ordinal model Rank association or predicted category probabilities across X <i>Delay minutes to satisfaction</i>
Ratio	Interval	Scatter / correlation / regression Scatterplot, fitted relationship and residual checks <i>Machine load to temperature</i>
Ratio	Ratio	Scatter / correlation / regression Scatterplot, fitted relationship and residual checks <i>Advertising spend to revenue</i>

Nominal predictor

Indicator codes with a reference; never average arbitrary labels.

Ordinal predictor

Keep order; indicators or justified trend score. Numeric labels do not prove equal gaps.

Interval predictor

Meaningful numerical increments; consider nonlinearity and scaling.

Ratio predictor

Meaningful numerical increments; consider nonlinearity and scaling.

For nominal Y, distinguish binary from unordered multicategory outcomes. For ordinal Y, preserve category order. For numerical Y, counts, bounded proportions, positive skew, censoring and dependence may call for a different family. Interval and ratio commonly share linear methods because regression models differences; a meaningful zero affects interpretation, not automatic normality.

Sources: [UCLA OARC \(n.d.\)](#); [statsmodels developers \(n.d.\)](#); [statsmodels developers \(n.d.\)](#); [Velleman & Wilkinson \(1993\)](#)

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Statistical method families

Each method card connects a question to an estimand, checks and useful visual evidence. “Beyond 1015” is optional exploration; it does not expand the assessment scope.

Compare numerical means

1015 core

Do independent groups have different mean delivery times?

Candidate methods: Two groups: Welch t-test is a useful default. Three or more: one-way ANOVA when its variance model fits; Welch ANOVA when group variances differ.

Report: Mean difference in the original units, with a confidence interval. For several groups, specify contrasts and multiplicity control.

Check: Independent observations; influential outliers; sample sizes and sampling model. Small-sample inference needs suitable distributional assumptions. Equal variance is not required by Welch.

Show: Group dots plus box plots; mean and interval plot. A box plot’s whiskers are not a confidence interval.

Sources: [SciPy developers \(n.d.\)](#); [SciPy developers \(n.d.\)](#)

Compare paired numerical measurements

1015 core

Did the same teams reduce turnaround time?

Candidate methods: Paired t-test analyzes within-unit differences. For an appropriate rank alternative, Wilcoxon signed-rank uses the numerical paired differences and a symmetric-difference assumption.

Report: Mean within-unit change and interval; report direction and units.

Check: Pairs must correspond to the same or matched units. Inspect the differences, not only the before and after marginal distributions. For purely ordinal labels, a sign or marginal-homogeneity approach can be more defensible than treating label gaps as numerical.

Show: Paired lines and a distribution of differences.

Sources: [UCLA OARC \(n.d.\)](#); [SciPy developers \(n.d.\)](#)

Compare ordered distributions

1015 connection

Do two service groups receive different satisfaction ratings?

Candidate methods: Independent groups: Mann-Whitney U (two) or Kruskal-Wallis (three or more). Repeated ordered responses can use Friedman; model the full category distribution when needed.

Report: Rank-based effect or category probabilities; report distributions. These are not automatically tests of medians.

Check: Independence for between-group tests; ties and small-sample inference. A location/median interpretation requires additional distribution-shape conditions. Repeated designs require a matched method.

Show: Ordered proportions, ECDFs, or jittered category points.

Sources: [SciPy developers \(n.d.\)](#); [UCLA OARC \(n.d.\)](#); [SciPy developers \(n.d.\)](#); [SciPy developers \(n.d.\)](#)

Compare categorical distributions

1015 core

Is purchase status associated with sales channel?

Candidate methods: Independent contingency table: chi-square independence when expected-count approximation is suitable; Fisher exact or an appropriate exact/Monte Carlo method for sparse tables.

Report: For 2×2 tables, risk difference, risk ratio or odds ratio with intervals. For larger tables, Cramér's V and category-specific proportions.

Check: Use counts, not percentages as observations. Inspect expected counts, sampling scheme and independent units. Matched binary outcomes need McNemar; its informative data are discordant pairs.

Show: Grouped/100% stacked bars, mosaic or table heatmap with clear denominator.

Sources: [SciPy developers \(n.d.\)](#); [statsmodels developers \(n.d.\)](#)

Measure numerical or rank association

1015 core

How does advertising spend vary with revenue?

Candidate methods: Pearson describes linear association of meaningful numerical values. Spearman or Kendall describes ordinal/monotonic association. A binary 0/1 variable with a numerical variable yields point-biserial Pearson correlation.

Report: Correlation and interval where appropriate; describe units and shape, including nonlinear patterns.

Check: Association is not causation. Check outliers, restricted ranges, confounding and independent units. Neither a large sample nor a small p-value resolves these issues.

Show: Scatterplot, with groups shown by color or facets; do not connect unrelated points.

Sources: [SciPy developers \(n.d.\)](#); [UCLA OARC \(n.d.\)](#); [SciPy developers \(n.d.\)](#)

Model a numerical outcome

1015 core

How does delivery time relate to order size after adjusting for channel?

Candidate methods: Linear regression with numerical predictors and indicator-coded categories; add interactions or flexible terms only when the question and data support them.

Report: Conditional coefficient in outcome units per predictor unit, confidence interval, and validation error for predictions.

Check: Check residual form, unequal variance, dependence, influence and collinearity. Normality concerns residuals for conventional small-sample inference, not a requirement that each predictor be normal. Robust standard errors do not repair a wrong model or clustering unless designed for it.

Show: Scatter/fitted line; effect plot; residual versus fitted, Q-Q and influence plots.

Sources: [NIST/SEMATECH \(n.d.\)](#); [UCLA OARC \(n.d.\)](#)

Model a binary outcome

Beyond 1015

How does customer tenure relate to cancellation probability?

Candidate methods: Binary logistic regression for individual events; a binomial model can also retain successes and trial counts.

Report: Predicted probability, probability difference, or odds ratio with interval. Odds ratios are not risk ratios.

Check: Correct coding, separation, sufficient events, calibration and dependence. Encode categorical predictors deliberately; check continuous-predictor functional form on the log-odds scale.

Show: Probability curve and intervals; calibration plot; threshold-specific confusion matrix.

Sources: [statsmodels developers \(n.d.\)](#)

Model an ordered categorical outcome

Beyond 1015

How do delivery delays relate to satisfaction categories?

Candidate methods: Ordinal logistic regression is a candidate. A proportional-odds model uses common slopes across category thresholds.

Report: Category probabilities and cumulative odds ratios, with a stated direction of ordering.

Check: Assess proportional odds and category sparsity. If this assumption fails, consider a more flexible ordinal model rather than silently ignoring it.

Show: Predicted category-probability curves or ordered stacked probabilities.

Sources: [statsmodels developers \(n.d.\)](#); [Liddell & Kruschke \(2018\)](#)

Model an unordered multicategory outcome

Beyond 1015

Which acquisition channel does a customer choose?

Candidate methods: Multinomial logistic regression with a reference category; do not impose an invented order.

Report: Predicted probabilities by category; contrasts against the stated reference.

Check: Sparse categories, meaningful reference, calibration, dependence and model-specific assumptions.

Show: Category probability plot; confusion matrix for classification decisions.

Sources: [UCLA OARC \(n.d.\)](#)

Model counts and exposure rates

Beyond 1015

Which stores receive more support tickets per operating hour?

Candidate methods: Poisson count regression with $\log(\text{exposure})$ offset for unequal observation windows. Negative binomial is a candidate for overdispersion; hurdle/zero-inflated models need a substantive two-process rationale.

Report: Expected counts or rates and rate ratios, with intervals.

Check: Integer counts; exposure > 0 ; independence or suitable cluster model; mean-variance relationship and zeros. Negative binomial does not fix dependence by itself.

Show: Discrete count distribution, exposure-adjusted rate plot and model checks.

Sources: [statsmodels developers \(n.d.\)](#)

Model proportions carefully

Beyond 1015

How does staffing relate to the share of a workday spent on analysis?

Candidate methods: Known successes out of trials: binomial model. Continuous proportions strictly in (0,1): beta regression. Continuous endpoints or other support require an appropriate extension or fractional-response approach.

Report: Predicted fraction, percentage-point difference and intervals; preserve denominators for counted fractions.

Check: Do not equate every percentage with a binomial experiment. Check how it was generated, bounds, endpoint values and repeated units.

Show: Bounded distribution and probability/fraction curves.

Sources: [Cribari-Neto & Zeileis \(2010\)](#); [statsmodels developers \(n.d.\)](#)

Analyze time and repeated observations

Beyond 1015

Will next month's orders exceed capacity?

Candidate methods: Trend/seasonal description; time-series models for forecasting. Longitudinal units may need mixed models or generalized estimating equations. Survival analysis addresses time to an event with censoring.

Report: Forecast plus prediction interval; within-unit effects or survival probabilities for the corresponding design.

Check: Time ordering, regularity, autocorrelation, censoring mechanism and leakage. Validate forecasts using later periods, not a randomly shuffled split.

Show: Time line, seasonal panels, autocorrelation, survival curve or longitudinal small multiples.

Sources: [Hyndman & Athanasopoulos \(2021\)](#); [Hyndman & Athanasopoulos \(2021\)](#); [UCLA OARC \(n.d.\)](#); [statsmodels developers \(n.d.\)](#); [statsmodels developers \(n.d.\)](#)

Estimate uncertainty and practical value

1015 core

Is the estimated improvement precise enough for a business decision?

Candidate methods: Choose an interval for the estimand and design. Bootstrap can help for suitable statistics; preserve pairing, clustering and time structure when resampling.

Report: Estimate, confidence level, interval, unit, sample size and practical threshold. SD describes observation spread; SE describes estimator precision.

Check: A 95% confidence procedure covers the fixed target in about 95% of repeated samples under its assumptions. A p-value is not the probability the null is true, and nonsignificance does not prove equivalence.

Show: Point-and-interval plot; forecast prediction bands for future observations, clearly labeled.

Sources: [Wasserstein & Lazar \(2016\)](#); [Altman & Bland \(2005\)](#); [statsmodels developers \(n.d.\)](#); [statsmodels developers \(n.d.\)](#); [NIST/SEMATECH \(n.d.\)](#)

Analyze many variables or constrained shares

Beyond 1015

Can we summarize several metrics or compare expense composition?

Candidate methods: PCA for numerical variation; factor analysis for an explicit latent-variable model. Log-ratio methods for compositional shares; circular methods for periodic directions.

Report: Loadings and explained variance; interpretable ratios or directional summaries for the appropriate question.

Check: Scaling, geometry, zeros, validation and business interpretation. Do not apply a linear mean to a circular value or treat constrained shares as independent.

Show: Correlation matrix, PCA biplot, ternary plot or polar histogram.

Sources: [scikit-learn developers \(n.d.\)](#); [van den Boogaart & Tolosana-Delgado \(n.d.\)](#); [SciPy developers \(n.d.\)](#)

07 / FIELD GUIDE

Choose the visual by purpose

Start with the message, then verify that the visual fits the values and the estimation task. A chart describes evidence; it is not automatically a statistical test.

Comparison

What differs across items or groups?

Bars/dots for counts and summaries; raw distributions and intervals for estimated effects.

Distribution

What values are common, rare or extreme?

Histogram/ECDF for numerical values; ordered bars for categories; box + points by group.

Relationship

How do measures vary together?

Scatterplot for two numerical measures; cross-tabs for categories; distributions across groups.

Composition

What makes up the whole?

Stacked bars for shares; waterfall for signed contribution; log-ratio methods for constrained vectors.

Change over time

What changes, repeats or drifts?

Line or seasonal panels; respect dependence. Chart choice alone does not provide a forecast.

Uncertainty

How precise is the estimate?

Point-and-interval plots; probability or prediction bands with the interval type named.

Diagnostics

Why should we trust this model?

Residual, Q-Q, influence and calibration views matched to the model family.

Sources: [Abela \(2006; updated 2020\)](#); [Holtz & Healy \(n.d.\)](#); [NIST/SEMATECH \(n.d.\)](#)

Bar / column

Comparison

Combination: Category + count or summary

Example: Channels: web 42, store 33, partner 25 orders.

Design check: Use a zero baseline for bar length; order categories intentionally. Horizontal bars help long labels.

Dot / lollipop

Comparison

Combination: Category + numerical summary

Example: Mean delay: web 18, store 12, partner 25 minutes.

Design check: Position is easy to compare. Label units; show uncertainty when means are estimates.

Point + interval

Uncertainty

Combination: Estimate + lower/upper limits

Example: Conversion estimate 12%, 95% Wilson interval 8.2% to 17.2%.

Design check: State confidence versus prediction interval. Overlap alone is not a formal test of a difference.

Box + raw points

Distribution

Combination: Category + numerical observations

Example: Order durations within three channels.

Design check: Explain median, quartiles and whisker rule. Outlier flags are review signals, not deletion instructions.

Histogram

Distribution

Combination: One numerical variable

Example: Delivery minutes grouped into 5-minute bins.

Design check: Bin choice can change the story. Bars touch; for unequal bins use density so area represents proportion.

ECDF

Distribution

Combination: One ordered numerical variable

Example: Fraction of orders delivered within each duration.

Design check: ECDF = empirical cumulative distribution function. It avoids bin choices and makes percentiles visible.

Density / violin

Distribution

Combination: Numerical values; optionally categories

Example: The shape of delay distributions by channel.

Design check: Smoothing adds assumptions and can hide small samples. Show sample size and points where feasible.

Scatterplot

Relationship

Combination: Two numerical measures

Example: Advertising spend versus revenue by store.

Design check: Each point is a unit. Show groups or facets to inspect confounding and outliers.

Bubble plot

Relationship

Combination: Two numerical positions + positive size

Example: Spend, revenue and number of customers.

Design check: Size circles by area, not radius; use a legend. A third variable is often clearer in facets.

Heatmap / mosaic

Relationship

Combination: Two categories + frequency/proportion

Example: Channel by purchase status.

Design check: Heatmap color shows magnitude; a mosaic uses area. State row/column denominator; do not average category codes.

Ordered / diverging bars

Composition

Combination: Ordered response categories + shares

Example: Five satisfaction categories within each channel.

Design check: Preserve category order. Separate positive and negative responses around a meaningful center.

100% stacked bars

Composition

Combination: Category + exhaustive shares

Example: 40% labor, 35% materials, 25% logistics.

Design check: Shows relative mix, hides total volume. Add totals or a companion chart when size matters.

Stacked area

Composition

Combination: Time + nonnegative component totals

Example: Sales by channel over six months.

Design check: Useful for total and broad mix. Interior bands lack a common baseline; use small multiples for precise comparisons.

Waterfall

Composition

Combination: Ordered signed contributions

Example: Opening 100, web +20, store -10, closing 110.

Design check: Contributions must reconcile. Distinguish totals from increments; avoid unexplained accumulation.

Line / small multiples

Change over time

Combination: Ordered time + numerical value

Example: Weekly orders across three channels.

Design check: Keep time spacing meaningful; show gaps. Many series are clearer in aligned panels.

Seasonal / polar plot

Change over time

Combination: Repeating time or direction + value

Example: Demand by hour or month; true wind direction.

Design check: Use polar geometry for a genuinely cyclic question. Regular seasonal panels often compare values more clearly.

Residual / Q-Q plot

Diagnostics

Combination: Fitted value, residual or quantile

Example: Residuals versus fitted delivery time.

Design check: Residual pattern checks form/spread; Q-Q checks tail/distribution behavior. A good-looking plot is not proof of every assumption.

Treemap / ternary

Composition

Combination: Positive hierarchy / exactly three shares

Example: Cost hierarchy; three budget shares summing to 100%.

Design check: Treemaps trade precise comparison for hierarchy. Ternary positions represent three constrained shares, not three independent axes.

Map

Geography

Combination: Spatial location + count, rate or category

Example: Tickets per 100 customer-hours by service area.

Design check: Use rates when populations differ; define missing areas and spatial units. A map is for geography, not a decorative substitute for comparison.

Correlation matrix / biplot

Multivariate

Combination: Several numerical or suitably encoded variables

Example: Relationships among spend, revenue, tenure and delay.

Design check: Correlation is pairwise; PCA biplots depend on scaling. Neither establishes causation or a validated latent construct.

Sources: Holtz & Healy (n.d.); NIST/SEMATECH (n.d.); NIST/SEMATECH (n.d.)

08 / FIELD GUIDE

Worked business examples

All data are synthetic teaching inputs. Calculations were checked independently with statistical libraries or explicit arithmetic. Interpret uncertainty under the stated design; never treat invented data as observed business evidence.

EXAMPLE 01 / Binary nominal → counted fraction

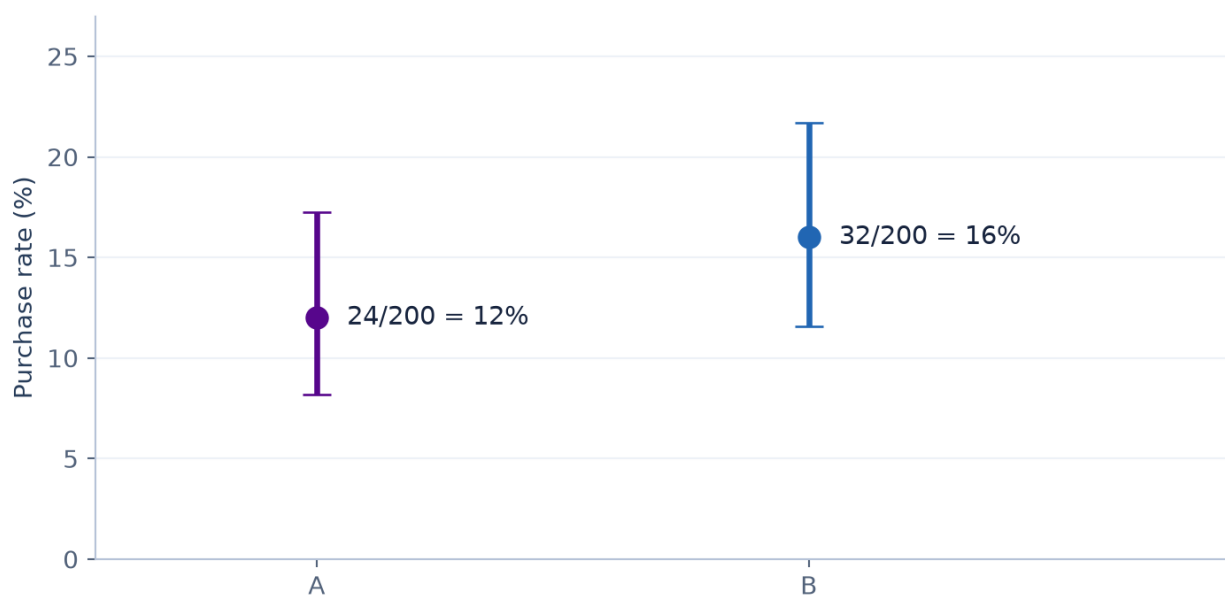
A purchase flag becomes a proportion

Did a new page improve conversion?

Inputs

- A: 24 purchases from 200 independently randomized eligible visitors.
- B: 32 purchases from 200 independently randomized eligible visitors.
- Synthetic teaching example; one visitor is one independent unit.

Conversion with 95% Wilson intervals



1. A rate = $24 / 200 = 12\%$; B rate = $32 / 200 = 16\%$.
2. Absolute difference = $16\% - 12\% = +4$ percentage points.
3. Relative lift = $(16\% - 12\%) / 12\% = 33.3\%$.
4. A 95% Wilson interval uses the binomial count and denominator; the difference uses a Newcombe interval from the two Wilson limits.

A: 12.0% (95% Wilson CI 8.2% to 17.2%). B - A: +4.0 points (95% Newcombe CI -2.9 to 10.9 points).

The estimated gain is positive, but the difference interval includes zero. It also spans gains that could matter in practice. This neither establishes an improvement nor proves no effect. Check the predeclared decision threshold, randomization and repeated-visitor rules.

Sources: [statsmodels developers \(n.d.\)](#); [statsmodels developers \(n.d.\)](#); [Wasserstein & Lazar \(2016\)](#)

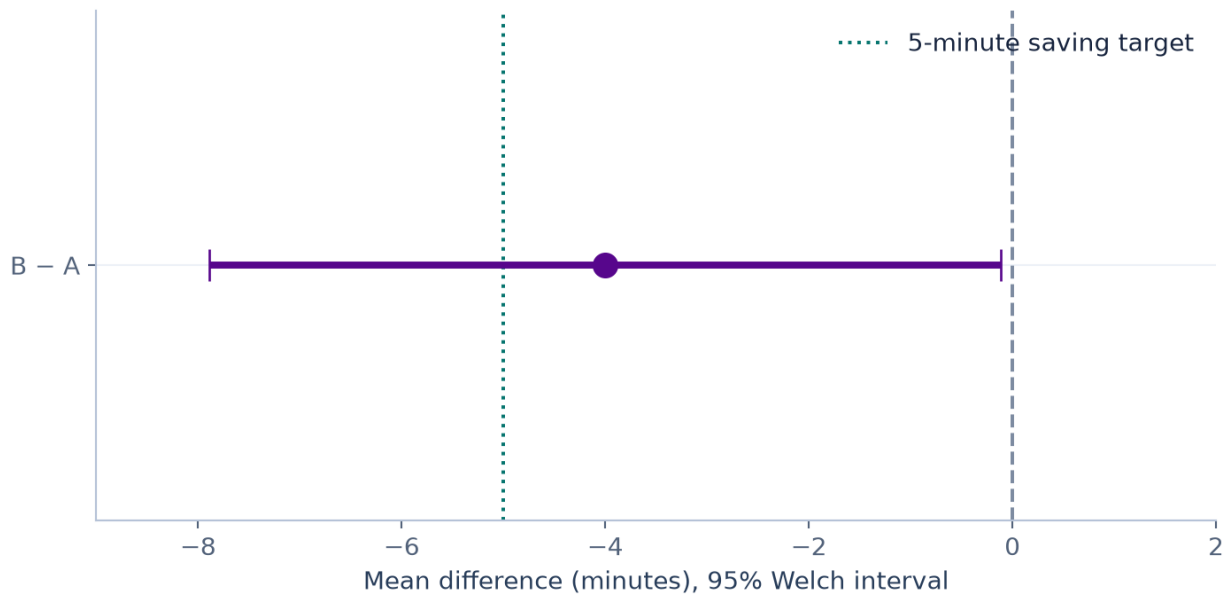
Course connections: Session 04, Session 05, Session 06

EXAMPLE 02 / Nominal predictor + ratio outcome**Channel groups and delivery minutes**

Are mean delivery times different?

Inputs

- Channel A: $n = 40$, mean = 32 minutes, sample SD = 8.
- Channel B: $n = 45$, mean = 28 minutes, sample SD = 10.
- Independent random samples; summary values are synthetic.

B minus A: mean delivery difference

1. Target: population mean for B minus population mean for A.
2. Estimate = $28 - 32 = -4$ minutes.
3. $SE = \sqrt{(8^2/40 + 10^2/45)}$. Use the Welch degrees of freedom. not a pooled-variance shortcut.

B - A = -4.00 minutes; SE = 1.955; df = 82.13; 95% CI -7.89 to -0.11 minutes; two-sided p = 0.0440.

The interval estimates how much lower B's population mean might be under the design assumptions. A business target of at least 5 minutes saved is not established by this interval. Do not claim causality from observational channels.

Sources: [SciPy developers \(n.d.\)](#)

Course connections: Session 05, Session 06, Session 08

EXAMPLE 03 / Paired ratio measurements

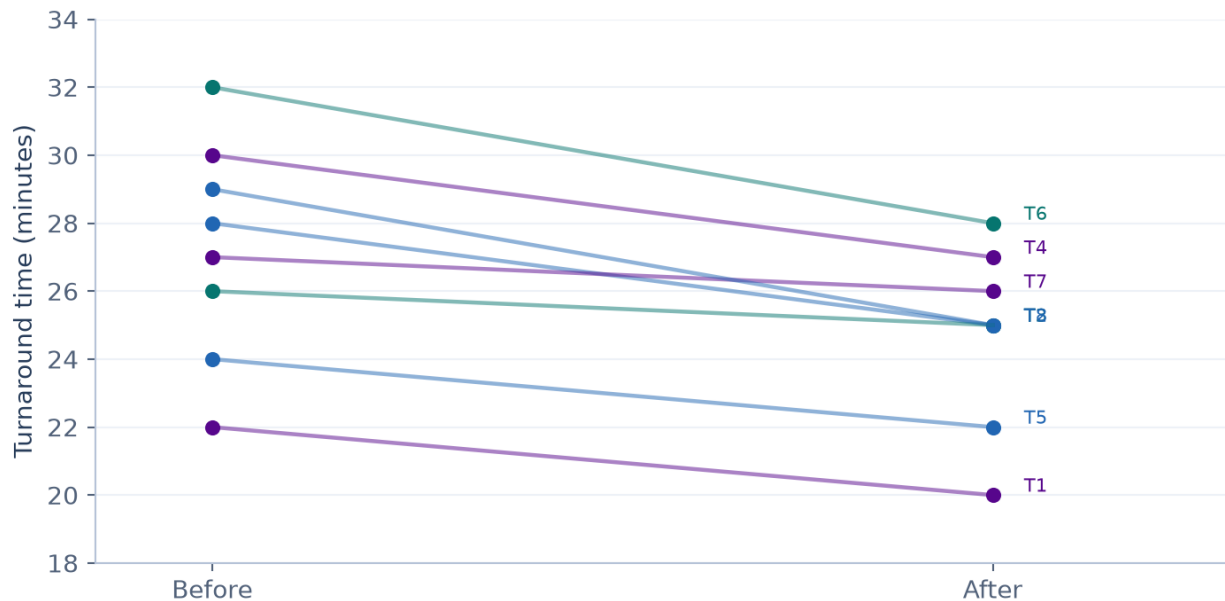
Same teams, two measurements

Did turnaround time change within teams?

Inputs

- Before (minutes): 22, 28, 26, 30, 24, 32, 27, 29.
- After (minutes): 20, 25, 25, 27, 22, 28, 26, 25.
- Each position is the same team; eight independent teams.

Eight teams: paired before and after



1. Compute each after - before change: -2, -3, -1, -3, -2, -4, -1, -4.
2. Mean change = -2.5 minutes. Analyze these eight differences with a paired t procedure.

Mean change = -2.50 minutes; 95% CI -3.50 to -1.50; two-sided p = 0.0006.

This small-sample interval relies on a suitable difference distribution and independent teams. The average within-team decrease is clear in these data; a before/after comparison alone does not isolate the intervention from other changes.

Sources: [UCLA OARC \(n.d.\)](#); [SciPy developers \(n.d.\)](#)

Course connections: Session 06, Session 08

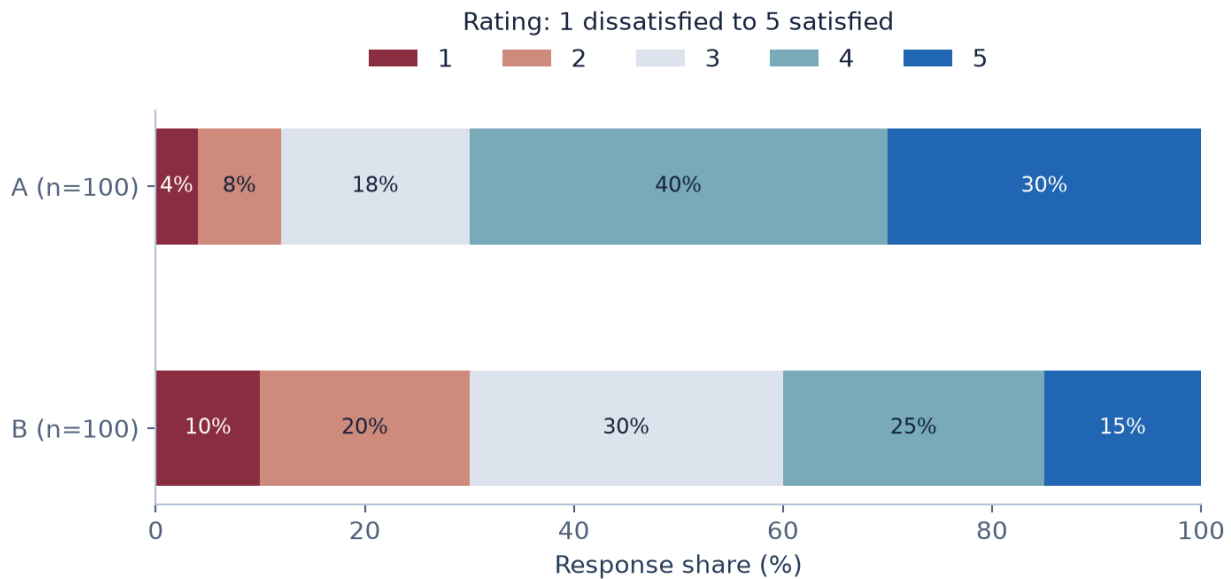
EXAMPLE 04 / Nominal group + single ordinal item**Keep the shape of a satisfaction scale**

How do the response distributions differ?

Inputs

- Ratings 1 to 5: very dissatisfied, dissatisfied, neutral, satisfied, very satisfied.
- Group A counts: 4, 8, 18, 40, 30. Group B: 10, 20, 30, 25, 15. Each n = 100.

Satisfaction: preserve all five categories



1. A satisfied/very satisfied share = $(40 + 30) / 100 = 70\%$.
2. B corresponding share = $(25 + 15) / 100 = 40\%$.
3. A median category is 4; B median category is 3. Retain all five categories in the chart.

A has 70% in categories 4 or 5; B has 40%, an observed 30-point difference. The full distribution shows where responses move.

The threshold summary loses detail and is not itself proof of a population effect. Predeclare the threshold if testing it. Use ordinal regression or a rank/distribution comparison for the full outcome; do not infer equal satisfaction gaps from the code numbers.

Sources: [Liddell & Kruschke \(2018\)](#); [statsmodels developers \(n.d.\)](#); [SciPy developers \(n.d.\)](#)

Course connections: Session 03, Session 08, Session 09

EXAMPLE 05 / Counted fractions + unequal denominators

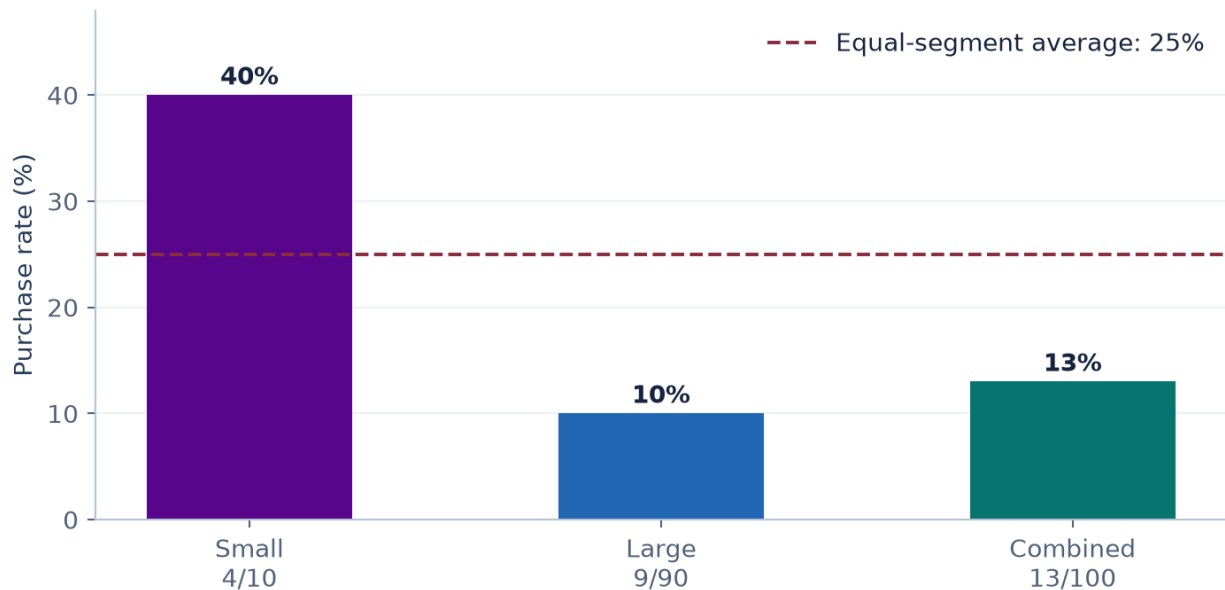
Why averaging percentages can fail

What is the overall purchase rate?

Inputs

- Small segment: 4 purchases / 10 visits = 40%.
- Large segment: 9 purchases / 90 visits = 10%.

The denominator changes the overall rate



1. Simple average of segment percentages = $(40\% + 10\%) / 2 = 25\%$.
2. Combined purchases / combined visits = $(4 + 9) / (10 + 90) = 13\%$.
3. Equivalent weighted mean = $(10 \times 40\% + 90 \times 10\%) / 100 = 13\%$.

The combined eligible-visit rate is 13%, not 25%.

The simple mean answers “what is the average segment rate with each segment weighted equally?” The weighted rate answers “what fraction of all visits purchased?” They are different estimands. Show segment volumes beside percentages.

Sources: [statsmodels developers \(n.d.\)](#); [Velleman & Wilkinson \(1993\)](#)

Course connections: Session 02, Session 03, Session 09

EXAMPLE 06 / Binary outcome + group + segment

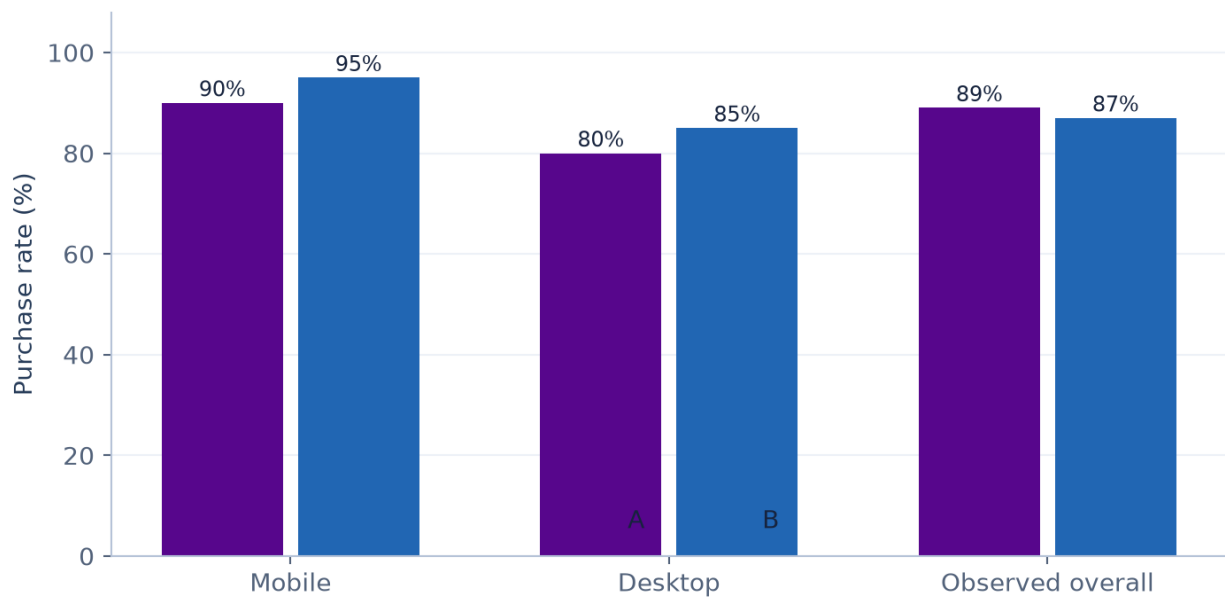
An aggregate comparison can reverse

Which page has the higher conversion rate?

Inputs

- A: mobile $81/90 = 90\%$; desktop $8/10 = 80\%$; overall $89/100 = 89\%$.
- B: mobile $19/20 = 95\%$; desktop $68/80 = 85\%$; overall $87/100 = 87\%$.

A mix reversal: subgroup versus aggregate



1. Overall, A is 2 points higher than B.
2. Within mobile, B is 5 points higher; within desktop, B is also 5 points higher.
3. A has 90% mobile visits; B has 20%. With equal 50/50 segment weights: $A = 85\%$. $B = 90\%$.

B is higher within each segment, while A is higher in the observed aggregate. The different segment mix explains the reversal.

This synthetic example demonstrates confounding by mix, not proof of a causal page effect. Compare like segments, define the target mix, and use a suitable stratified or adjusted analysis with uncertainty.

Sources: [SciPy developers \(n.d.\)](#); [UCLA OARC \(n.d.\)](#)

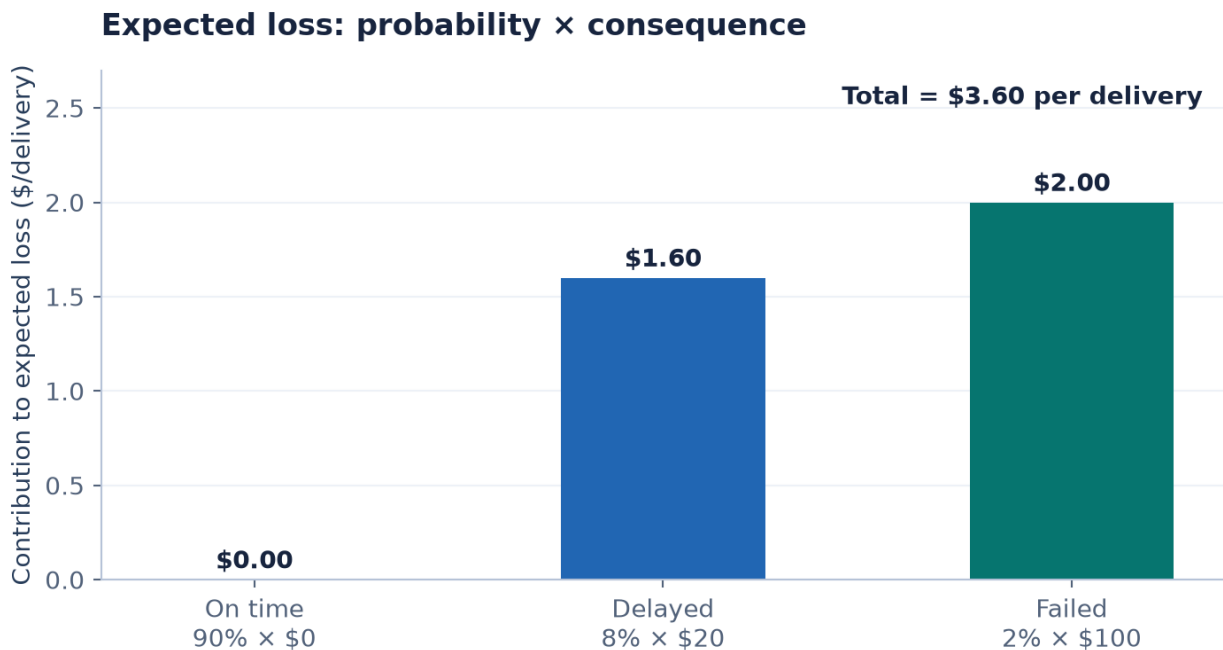
Course connections: Session 02, Session 09, Session 10

EXAMPLE 07 / Nominal scenarios + probabilities + ratio loss**Risk decomposes into cases and consequences**

What is the expected loss of one delivery?

Inputs

- Mutually exclusive scenarios: on-time probability 0.90, loss \$0; delayed 0.08, loss \$20; failed 0.02, loss \$100.
- Probabilities sum to 1. These are assumed teaching inputs, not observed estimates.



1. Expected loss = $0.90 \times \$0 + 0.08 \times \$20 + 0.02 \times \$100$.

2. Expected loss = $\$1.60 + \$2.00 = \$3.60$ per delivery.

Expected loss is \$3.60 per delivery. This is an average over repeated comparable deliveries, not the loss of any one delivery.

Show both probability and consequence: the rare failed delivery contributes \$2.00 of the \$3.60 expectation. Simulation can explore scenarios but cannot repair incorrect probabilities or loss definitions.

Sources: [statsmodels developers \(n.d.\)](#)

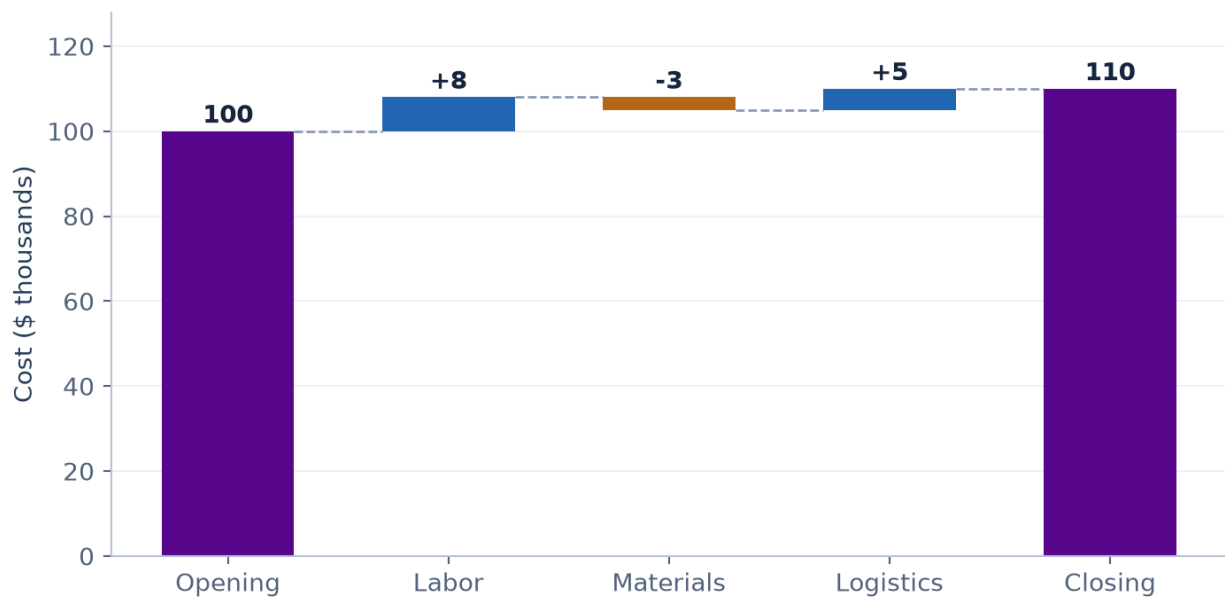
Course connections: Session 04

EXAMPLE 08 / Nominal components + numerical amounts**Parts, totals and a signed reconciliation**

What contributes to the cost total and its change?

Inputs

- Current cost: labor \$40k, materials \$35k, logistics \$25k; total \$100k.
- Next period changes: labor +\$8k, materials -\$3k, logistics +\$5k.

Reconcile the total cost change

1. Current shares: 40%, 35%, 25%, summing to 100%.
2. New total = \$100k + \$8k - \$3k + \$5k = \$110k.
3. New amounts: \$48k. \$32k. \$30k: new shares: 43.6%. 29.1%. 27.3%.

The total grows by \$10k (10%). Labor grows by \$8k and its share rises by 3.6 percentage points.

Use a waterfall for the signed change and stacked bars for the share mix. Percent change, percentage-point change and contribution to total are different quantities. Shares are linked by the fixed-sum constraint.

Sources: [van den Boogaart & Tolosana-Delgado \(n.d.\)](#); [Abela \(2006; updated 2020\)](#)

Course connections: Session 02, Session 03, Session 13

09 / FIELD GUIDE

Design and assumption checks

A method should answer the target question under a defensible design. Use diagnostics as evidence about a model, not as a ritual that guarantees correctness.

Independence is a design fact

Repeated visits from one customer are not automatically independent visits. Use unit-level aggregation or a model/SE that respects the dependency. Pairing and clustering cannot be fixed by choosing a prettier chart.

Choose the estimand before the test

Estimand means the population quantity being targeted: mean difference, risk difference, odds ratio, rate ratio, median or category probabilities. Rank tests answer a distribution question unless extra assumptions support a location interpretation.

Normality belongs to the right object

A paired t-test concerns differences; regression inference concerns errors/residuals under the model. Numerical predictors need not each be normally distributed. Skew, extreme outliers, sample size and design affect robustness.

Missing and zero are different

Missing, not applicable, censored, structural zero and observed zero need distinct rules. Do not convert every blank to zero, and never remove outliers solely to obtain significance.

Expected counts and sparse categories

A chi-square approximation depends on expected cell counts, not simply a large total sample. Use an appropriate exact or simulation-based method if sparse, and retain the sampling design.

Multiple comparisons need a plan

Predefine primary comparisons. Control family-wise error or false discoveries for the relevant family, and report the procedure. Looking repeatedly at an A/B result requires a sequential plan; ordinary fixed-horizon p-values are not a stopping rule.

Prediction needs honest evaluation

Separate fitting and evaluation at the correct unit and time. Report MAE/RMSE for numerical predictions; calibration and decision-specific errors for probabilities. A high R^2 on training data is not evidence of deployment performance.

Fairness and confounding need context

Inspect both rates and denominators by relevant groups, with uncertainty and lawful/ethical use of attributes. An adjusted regression association does not by itself remove unmeasured confounding or establish fairness.

Sources: [NIST/SEMATECH \(n.d.\)](#); [Wasserstein & Lazar \(2016\)](#); [SciPy developers \(n.d.\)](#); [Hyndman & Athanasopoulos \(2021\)](#); [UCLA OARC \(n.d.\)](#)

10 / FIELD GUIDE

All 14 GC1015 sessions

Curricular alignment uses the approved Fall 2026 GC1015-102 syllabus. These are guide connections, not claims about exact taught slides or PDF pages. Current Brightspace instructions govern dates, pacing and assessment conditions.

Session 01 / Orientation; readiness; business-question framing

M00 + M01

Define the decision, population, outcome, unit and units.

[Open the corresponding guide section](#)

Sources: [UCLA OARC](#) (n.d.)

Session 02 / KPIs, variables, grain, chart and data-story critique

M01 + M02

Translate KPI → operational definition → variable subtype → chart purpose.

[Open the corresponding guide section](#)

Sources: [Abela](#) (2006; updated 2020)

Session 03 / Data quality, distributions, descriptive evidence

M02

Audit missing values and duplicates; compare center, spread and distributions.

[Open the corresponding guide section](#)

Sources: [NIST/SEMATECH](#) (n.d.)

Session 04 / Conditional risk, expected value, simulation

M03

Separate event probabilities, counts and amounts; build conditional risk tables and expected-value scenarios.

[Open the corresponding guide section](#)

Sources: [statsmodels developers](#) (n.d.)

Session 05 / Sampling, bias, confidence intervals, precision

M04

Match interval to mean/proportion and design; distinguish SD from SE.

[Open the corresponding guide section](#)

Sources: [Altman & Bland](#) (2005)

Session 06 / Hypothesis tests, effects, practical significance, A/B guardrails

M05

Predefine the outcome, contrast, practical threshold and multiple-testing rule.

[Open the corresponding guide section](#)

Sources: [Wasserstein & Lazar \(2016\)](#)

Session 07 / Midterm Decision Analysis

M01-M05

Integrate question, grain, data quality, risk and inference. This guide does not change assessment rules.

[Open the corresponding guide section](#)

Sources: [Ezekwem \(2026\)](#)

Session 08 / Test selection, comparisons, assumptions, effects

M06

Use the combination map, then check independence, pairing and numerical support.

[Open the corresponding guide section](#)

Sources: [UCLA OARC \(n.d.\)](#)

Session 09 / Correlation, categorical association, confounding

M07

Compare scatter/rank/cross-tab evidence; inspect subgroup structure and denominators.

[Open the corresponding guide section](#)

Sources: [SciPy developers \(n.d.\)](#)

Session 10 / Regression coefficients, conditional relationships, model plan

M08

Encode predictors deliberately; state outcome family and conditional interpretation.

[Open the corresponding guide section](#)

Sources: [NIST/SEMATECH \(n.d.\)](#)

Session 11 / Diagnostics, prediction error, fairness, responsible use

M09

Check residuals/calibration, validate on unseen units, and inspect subgroup performance.

[Open the corresponding guide section](#)

Sources: [Hyndman & Athanasopoulos \(2021\)](#)

Session 12 / Focused project revision, reproducibility, QA clinic

M10

Freeze definitions, data versions, analysis choices and an evidence manifest.

[Open the corresponding guide section](#)

Sources: [Ezekwem \(2026\)](#)

Session 13 / Project rehearsal, executive communication, defense

M11

Choose one decision-focused visual, quantify uncertainty and explain limitations.

[Open the corresponding guide section](#)

Sources: [Holtz & Healy \(n.d.\)](#)

Session 14 / Final project showcase and oral verification

M01-M11

Defend measurement, method, chart and practical recommendation with reproducible evidence.

[Open the corresponding guide section](#)

Sources: [Ezekwem \(2026\)](#)

The calendar exception is not Session 07: the approved plan has no regular Wednesday class on October 14, 2026. This guide creates no replacement activity.

Sources: [Ezekwem \(2026\)](#)

11 / FIELD GUIDE

Beyond the four levels

Use these extensions when the question needs them. Explain the additional assumptions and keep the method connected to a decision.

Design interactions, not just pairs

Nominal $\times$ numerical predictors with a numerical outcome: test whether the slope of order size differs by channel using an interaction. Numerical $\times$ numerical predictors: show a response surface as a heatmap or contour, with observed-data support. Three categories: stratified tables or faceted proportions are often clearer than a 3D chart.

Sources: [UCLA OARC \(n.d.\)](#); [NIST/SEMATECH \(n.d.\)](#)

When two variables are not enough

Two numerical variables plus an ordered category can use a scatterplot faceted by the category. Two categories plus a count can use a table heatmap. Time plus category plus amount can use small-multiple lines. A fourth variable usually belongs in facets or separate panels rather than another visual encoding.

Sources: [Holtz & Healy \(n.d.\)](#); [NIST/SEMATECH \(n.d.\)](#)

Transformations preserve some meanings and change others

Binning a duration into short/medium/long creates an ordinal variable and loses detail. A z-score is a standardized difference with a chosen zero, not a ratio measure. A log of a positive amount turns multiplicative ratios into additive differences. Standardizing or taking logs does not make a variable normally distributed by definition.

Sources: [Velleman & Wilkinson \(1993\)](#); [statsmodels developers \(n.d.\)](#)

Use the right average

Arithmetic means summarize additive amounts. A geometric mean describes multiplicative factors when values are positive; +10% then -10% multiplies $1.10 \times 0.90 = 0.99$, a net -1%, even though the arithmetic average return is 0%. Harmonic means suit rates only under the relevant equal-distance/equal-work weighting. Denominator-weighted means are essential for pooled fractions. Coefficient of variation needs a meaningful zero and a stable positive mean.

Sources: [NIST \(2016\)](#); [Velleman & Wilkinson \(1993\)](#)

Alternative typologies explain the missing detail

Mosteller and Tukey's seven-part scheme distinguishes names, grades, ranks, counted fractions, counts, amounts and balances. Chrisman expands the framework for membership, log intervals, extensive and derived ratios, cyclic values, counts and absolute quantities. These are perspectives on data structure, not an additional automatic test selector. The useful lesson is to inspect support, geometry and how the variable was constructed.

Sources: [Wikipedia contributors \(2026\)](#); [Chrisman \(1998\)](#)

Beyond 1015 does not mean required

Explore logistic/ordinal/count models, survival, mixed models, time series, compositional analysis and PCA after mastering the course's descriptive, inferential, association and regression foundations. These links explain a progression, not additional assessed requirements. The approved assessment instructions determine what references and assistance are permitted.

Sources: [Ezekwem \(2026\)](#); [UCLA OARC \(n.d.\)](#)

Software pathways

Excel 365

Use explicit tables and units, counts/percentages, descriptive summaries, well-labeled charts and supported tests. Record formulas and the independent/paired choice. Excel does not choose the correct statistical model for you.

[Official learning / documentation](#)

SPSS

Set categorical value labels and missing codes deliberately. SPSS "Scale" groups interval and ratio variables; that software label does not remove the distinction. Check the selected procedure's design and assumptions.

[Official learning / documentation](#)

Python (optional)

SciPy supports descriptive and classical tests; statsmodels supports regression and many outcome families. Match the model to the data-generating process and preserve versions.

[Official learning / documentation](#)

R (optional)

Use factors and ordered factors deliberately, inspect formula coding, and record the analysis. Base statistics and specialist packages provide methods for many designs.

[Official learning / documentation](#)

12 / FIELD GUIDE

A modern reading of the chart chooser

The supplied chart is Andrew Abela's 2006 "Chart Suggestions: A Thought-Starter." It asks a good first question: what do you want to show? This guide builds original decision diagrams and adds estimation, dependence and model checks.

Keep the four original purposes

Abela's comparison, relationship, distribution and composition remain useful. This guide adds time, uncertainty and diagnostics so estimation and model checking are visible.

Use sample size to control detail, not invent a chart rule

A tiny numerical sample often deserves visible dots. Histograms need informative bins; a smooth density is not a "line histogram," and more observations do not automatically justify smoothing.

Use 2D evidence by default

Three variables often fit a scatterplot with color, size or facets. A 3D area chart can hide values through perspective and occlusion; use a heatmap or contour for a numerical response surface when appropriate.

Treat cyclic data as geometry

A radar chart for arbitrary categories is not a substitute for analyzing periodic time or direction. Use seasonal panels or circular summaries when the data wrap around.

Show both share and scale when needed

A 100% stacked chart hides volume; add totals. A pie can communicate a simple few-part whole, but bars are usually easier for precise comparison. Avoid 3D pies and many thin slices.

Keep encodings honest

Marimekko/variable-width columns combine two quantities through area and need clear labels. Bubble size should encode area. Never use decorative perspective or a truncated bar baseline to exaggerate an effect.

Sources: [Abela \(2006; updated 2020\)](#); [Holtz & Healy \(n.d.\)](#); [NIST/SEMATECH \(n.d.\)](#); [SciPy developers \(n.d.\)](#)

13 / FIELD GUIDE

Report evidence that supports a decision

- 01.** State the decision, target population, comparison and practical threshold.
- 02.** Provide a data dictionary: grain, level, subtype, unit, valid values, missing codes, numerator/denominator and time window.
- 03.** Show sample sizes or exposure, exclusions, uncertainty method, and checks of independence or pairing.
- 04.** Name the statistical method, estimand, assumptions, effect and interval. Keep practical significance separate from a p-value.
- 05.** Use a chart with title, labeled units, readable categories, denominator and accessible color/shape cues. Describe the main pattern in text.
- 06.** For a model, report encoding, validation design, relevant error, diagnostics, and limits of causal interpretation.
- 07.** Save the source/version, transformations, analysis steps, software version and results so another analyst can reproduce them.
- 08.** End with an actionable recommendation, uncertainty that matters, and what evidence would change the decision.

A compact evidence sentence

“In the independent synthetic samples, Channel B averaged 4.0 fewer delivery minutes than A (95% Welch CI: 0.11 to 7.89 fewer). The estimate is useful, but the data do not establish a minimum 5-minute saving or a causal channel effect.”

Check and bring back: one defensible measurement definition, one candidate method with its assumptions, one labeled chart, and one decision statement with uncertainty. Use the current assignment’s instructions for what must be submitted.

14 / FIELD GUIDE

Glossary

Grain: What one row represents.

Estimand: The population quantity you want to estimate.

Parameter: A feature of the target population or model.

Statistic: A quantity calculated from observed sample data.

Effect size: Magnitude of a comparison or relationship, in interpretable units or a defined scale.

Confounding: Mixing a relationship with another variable that influences the comparison.

Residual: Observed outcome minus the value fitted by a model.

Exposure: Time, population or opportunities over which a count can occur.

Offset: A known term such as $\log(\text{exposure})$ with coefficient fixed at 1 in a count model.

Calibration: Agreement between predicted probabilities and observed event frequencies.

Censoring: The event time is only partly observed, such as knowing it exceeds a follow-up period.

ECDF: Fraction of observed values at or below each value.

Confidence interval: An interval from a procedure with stated repeated-sampling coverage for a target parameter.

Prediction interval: An interval for a future observation, including future-outcome uncertainty.

Percentage point: An absolute change between two percentages: 12% to 15% is +3 points.

Percent change: Relative change: 12% to 15% is $(15-12)/12 = 25\%$.

Odds: Probability divided by one minus probability.

Overdispersion: More outcome variability than the chosen model's variance assumption implies.

Interaction: A relationship whose size or direction depends on another predictor.

Compositional constraint: Parts must sum to a total, so their relative movements are linked.

15 / FIELD GUIDE

References and source notes

Research checked October 3, 2026. Author-date links appear throughout. Primary research, official statistical documentation and original visualization sources inform the synthesis. Wikipedia supplied orientation and taxonomy leads. Access notes distinguish documentation, abstracts and indexed excerpts.

Abela (2006; updated 2020). Choosing a good chart. [Read source.](#)

Source of the user-supplied chart. This guide uses original diagrams and modern recommendations; it does not reproduce the chart.

Altman & Bland (2005). Standard deviations and standard errors. [Read source.](#)

Primary Statistics Note reviewed in indexed excerpt

Chrisman (1998). Rethinking levels of measurement for cartography. [Read source.](#)

Publisher abstract reviewed; expanded taxonomy cross-checked with Wikipedia, not full paper

Cribari-Neto & Zeileis (2010). Beta regression in R. [Read source.](#)

Full web documentation reviewed

Ezekwem (2026). GC1015-102 approved Fall 2026 syllabus: Course Outline.

Local instructor-approved syllabus. All 14 session titles cross-checked. Links in the course map are guide sections, not verified slide or PDF page references. Brightspace provides current operational details.

Holtz & Healy (n.d.). From Data to Viz. [Read source.](#)

Full web documentation reviewed

Hyndman & Athanasopoulos (2021). Evaluating point forecast accuracy, §5.8. [Read source.](#)

Full web documentation reviewed

Hyndman & Athanasopoulos (2021). Forecasting: Principles and Practice, 3rd ed., Chapter 3. [Read source.](#)

Chapter introduction reviewed; further reading for decomposition

Liddell & Kruschke (2018). Analyzing ordinal data with metric models: What could possibly go wrong?. [Read source.](#)

Publisher abstract reviewed; not a full-paper methods review

NIST (2016). NIST IR 8101, §2.3: Scales. [Read source.](#)

Scale definitions reviewed in indexed primary report excerpt

NIST/SEMATECH (n.d.). Confidence limits for the mean. [Read source.](#)

Primary documentation reviewed

NIST/SEMATECH (n.d.). Gallery of graphical techniques. [Read source.](#)

Full web documentation reviewed

NIST/SEMATECH (n.d.). Histogram. [Read source.](#)

Full web documentation reviewed

NIST/SEMATECH (n.d.). How can I tell if a model fits my data?. [Read source.](#)

Full web documentation reviewed

SciPy developers (n.d.). Friedman repeated-samples test. [Read source.](#)

Primary documentation reviewed

SciPy developers (n.d.). Kruskal-Wallis H test. [Read source.](#)

Primary documentation reviewed

SciPy developers (n.d.). Pearson correlation. [Read source.](#)

Primary documentation reviewed

SciPy developers (n.d.). chi2_contingency: categorical independence. [Read source.](#)

Full web documentation reviewed

SciPy developers (n.d.). circmean: circular mean. [Read source.](#)

Full web documentation reviewed

SciPy developers (n.d.). f_oneway: ordinary and Welch ANOVA. [Read source.](#)

Full web documentation reviewed

SciPy developers (n.d.). mannwhitneyu: independent rank comparison. [Read source.](#)

Full web documentation reviewed

SciPy developers (n.d.). spearmanr: rank association. [Read source.](#)

Full web documentation reviewed

SciPy developers (n.d.). ttest_ind: independent means and Welch test. [Read source.](#)

Full web documentation reviewed

SciPy developers (n.d.). wilcoxon: paired signed-rank comparison. [Read source.](#)

Full web documentation reviewed

Stevens (1946). On the theory of scales of measurement. [Read source.](#)

Original framework; bibliographic metadata verified through Wikipedia and NIST. Full original article not reviewed.

UCLA OARC (n.d.). Choosing the correct statistical test. [Read source.](#)

Full web documentation reviewed

Velleman & Wilkinson (1993). Nominal, ordinal, interval, and ratio typologies are misleading. [Read source.](#)

Publisher abstract reviewed; full paper available through university-hosted manuscript

Wasserstein & Lazar (2016). The ASA statement on p-values: Context, process, and purpose. [Read source.](#)

Full web documentation reviewed

Wikipedia contributors (2026). Level of measurement. [Read source.](#)

Orientation source reviewed in Chrome and web browser. Not the sole authority for method selection.

scikit-learn developers (n.d.). Principal component analysis. [Read source.](#)

Full web documentation reviewed

statsmodels developers (n.d.). Confidence intervals for a binomial proportion. [Read source.](#)

Full web documentation reviewed

statsmodels developers (n.d.). Confidence intervals for two independent proportions. [Read source.](#)

Full web documentation reviewed

statsmodels developers (n.d.). Generalized estimating equations. [Read source.](#)

Primary documentation reviewed

statsmodels developers (n.d.). Generalized linear models. [Read source.](#)

Full web documentation reviewed

statsmodels developers (n.d.). McNemar test for paired binary observations. [Read source.](#)

Full web documentation reviewed

statsmodels developers (n.d.). Ordinal regression. [Read source.](#)

Full web documentation reviewed

statsmodels developers (n.d.). Survival and duration analysis. [Read source.](#)

Primary documentation reviewed

van den Boogaart & Tolosana-Delgado (n.d.). Compositions: author-maintained R package and book resources. [Read source.](#)

Author resource inspected through search index; further reading

Original synthesis and diagrams prepared for Professor Ken Ezekwem. No student data, assessment responses, instructor answer keys, or licensed chart reproduction are included. The guide is a learning reference; software and sources do not substitute for validating a particular analysis.